

A PRACTICAL GUIDE

Gamma & Beta

Two functions. A reusable method for definite integrals.

Gamma packages a power multiplied by exponential decay. Beta packages a power at each end of an interval. A good substitution brings many apparently unrelated integrals into one of these two forms.

The method: check the endpoints; choose a substitution; transform the differential and bounds; match the exponents; simplify the Gamma or Beta value.

The two definitions

$$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt, \quad s > 0.$$

$$B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt, \quad a, b > 0.$$

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

The first move often decides the problem

What you see	What to try
e^{-cx^r} on $[0, \infty)$	Set $t = cx^r$; look for Gamma.
x and $L - x$ on $[0, L]$	Set $x = Lt$; look for Beta.
$(1 + cx^r)^{-q}$ on $[0, \infty)$	Set $t = cx^r$; use the infinite-interval Beta form.
Powers of sine and cosine	Set $t = \sin^2 \theta$ on $[0, \pi/2]$.
$-\ln x$ on $[0, 1]$	Set $t = -\ln x$; the logarithm becomes a power.

How to use this guide

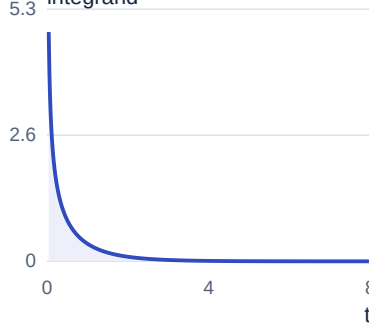
Pages 2–3 develop the functions and their connection. Pages 4–8 solve ten integrals. Page 9 collects the substitution formulas and their conditions. Page 10 gives eight practice problems; pages 11–12 give solutions and sources. All parameters are real unless explicitly stated otherwise. The prerequisites are substitution, integration by parts, and basic improper integrals. These methods evaluate *definite* integrals without requiring elementary antiderivatives.

Factorials between the integers

At zero, $t^{s-1}e^{-t}$ behaves like t^{s-1} , which has a finite integral precisely when $s > 0$. At infinity, the exponential controls every fixed power. Thus the defining integral converges for every real $s > 0$.

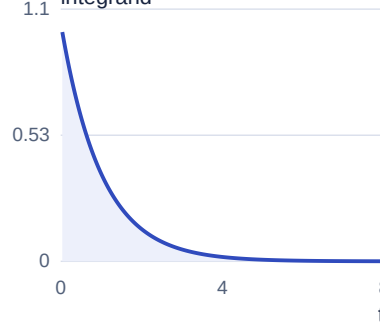
s = 0.5

Full integral = 1.7725
5.3 integrand



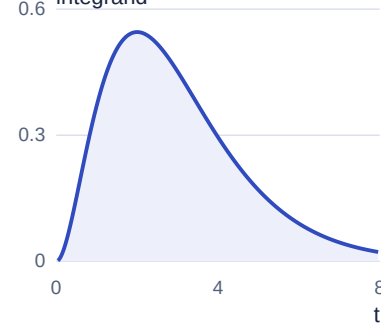
s = 1

Full integral = 1
1.1 integrand



s = 3

Full integral = 2
0.6 integrand



Each panel shows the unnormalized integrand for a different s , with its own vertical scale. Sampled window $0.04 \leq t \leq 7.96$; the listed value includes the omitted endpoint region and infinite tail. A divergent height at zero need not imply divergent area.

Integration by parts creates the factorial recurrence

Take $u = t^s$ and $dv = e^{-t} dt$. For $s > 0$, both endpoint terms vanish:

$$\Gamma(s+1) = [-t^s e^{-t}]_0^\infty + s \int_0^\infty t^{s-1} e^{-t} dt = s\Gamma(s).$$

Since $\Gamma(1) = 1$, repeated recurrence gives $\Gamma(n+1) = n!$ for integers $n \geq 0$. Remember the shift: $\Gamma(4) = 3! = 6$, not $4!$.

Why the half-integer values contain $\sqrt{\pi}$

Set $t = u^2$ in $\Gamma(1/2)$, giving $\Gamma(1/2) = 2J$ with $J = \int_0^\infty e^{-u^2} du$. Squaring and using polar coordinates in the first quadrant gives

$$J^2 = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta = \frac{\pi}{2} \cdot \frac{1}{2} = \frac{\pi}{4}.$$

The integrand is nonnegative, and $J > 0$. Hence $J = \sqrt{\pi}/2$ and $\Gamma(1/2) = \sqrt{\pi}$.

$$\Gamma(3/2) = \frac{\sqrt{\pi}}{2}, \quad \Gamma(5/2) = \frac{3\sqrt{\pi}}{4}, \quad \Gamma(7/2) = \frac{15\sqrt{\pi}}{8}.$$

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{4^n n!} \sqrt{\pi}, \quad n = 0, 1, 2, \dots$$

An integral is more than a recurrence. Many functions can interpolate factorial values. The Gamma integral specifies one particular extension; recurrence alone does not uniquely determine it.

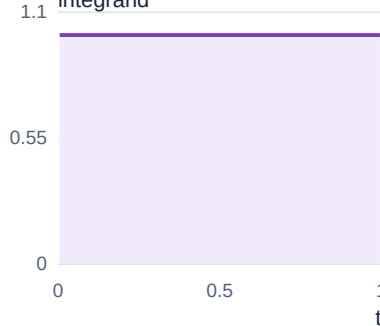
Standard identities: NIST DLMF §5.2, §5.4, §5.5.

Separate total size from proportion

In $B(a, b)$, the exponent $a - 1$ governs the endpoint at zero and $b - 1$ governs the endpoint at one. Both must exceed -1 . The substitution $t \mapsto 1 - t$ reflects the graph and proves $B(a, b) = B(b, a)$.

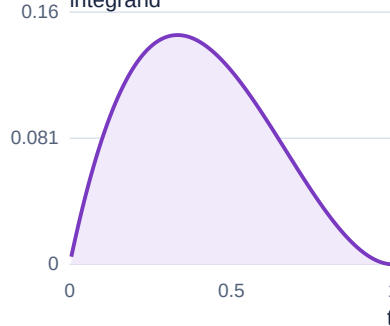
a = 1, b = 1

Full integral = 1
integrand



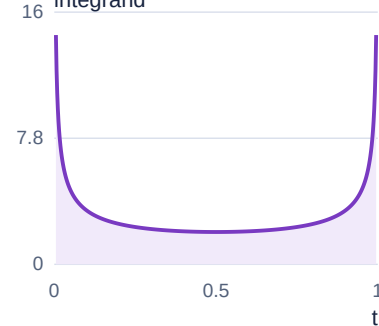
a = 2, b = 3

Full integral = 0.083333
integrand



a = 0.5, b = 0.5

Full integral = 3.1416
integrand



Unnormalized Beta integrands, each with its own vertical scale. The plotted window is $0.005 \leq t \leq 0.995$; listed areas include the omitted endpoint regions. Dividing the integrand by $B(a, b)$ produces a probability density with area 1.

Derive the Gamma–Beta relation

For $a, b > 0$, multiply the two Gamma integrals. Nonnegativity allows us to write a double integral by Tonelli's theorem:

$$\Gamma(a)\Gamma(b) = \int_0^\infty \int_0^\infty u^{a-1} v^{b-1} e^{-(u+v)} du dv.$$

Now set $r = u + v$ and $t = u/(u + v)$. Then $u = rt$, $v = r(1 - t)$, $r > 0$, and $0 < t < 1$. The absolute Jacobian is

$$\left| \det \begin{pmatrix} t & r \\ 1-t & -r \end{pmatrix} \right| = r.$$

Substitution separates the two variables:

$$\begin{aligned} \Gamma(a)\Gamma(b) &= \int_0^\infty \int_0^1 r^{a+b-1} e^{-r} t^{a-1} (1-t)^{b-1} dt dr \\ &= \Gamma(a+b)B(a, b). \end{aligned}$$

Divide by $\Gamma(a+b) > 0$. The variable r records total size; t records the fraction allocated to u .

Two useful consequences

$$B(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!} \quad (m, n \text{ positive integers}).$$

With $t = u/(1+u)$ and $dt = du/(1+u)^2$, the unit interval becomes the half-line:

$$B(a, b) = \int_0^\infty \frac{u^{a-1}}{(1+u)^{a+b}} du, \quad a, b > 0.$$

Standard Beta forms and relation: NIST DLMF §5.12.1–3.

WORKED INTEGRALS

Powers with exponential decay

The reusable identity is $\int_0^\infty x^{p-1} e^{-cx^r} dx = \Gamma(p/r)/(rc^{p/r})$ for $p, c, r > 0$. It follows from $t = cx^r$.

01. Scale an exponential

$$I = \int_0^\infty x^3 e^{-2x} dx$$

Convergence. Near zero the integrand behaves like x^3 ; exponential decay controls infinity.

Choose the substitution. Both limits stay 0 and infinity. Transform the differential as well as the power.

$$t = 2x, \quad x = t/2, \quad dx = dt/2$$

Match the exponent. The Gamma argument is one more than the exponent: t^3 means $s = 4$.

$$I = \frac{1}{16} \int_0^\infty t^3 e^{-t} dt = \frac{1}{16} \Gamma(4)$$

Evaluate. The factor $1/16$ contains three powers from x^3 and one from dx .

$$I = \frac{3!}{16} = \boxed{\frac{3}{8}}$$

02. Absorb a power in the exponential

$$I = \int_0^\infty \sqrt{x} e^{-3x^2} dx$$

Convergence. The power $x^{1/2}$ is integrable at zero; the exponential controls infinity.

Choose the substitution. A power inside the exponential suggests making that entire power the new variable.

$$t = 3x^2, \quad dx = \frac{t^{-1/2}}{2\sqrt{3}} dt$$

Combine the powers. Since $\sqrt{x} = (t/3)^{1/4}$, combining it with dx leaves $t^{-1/4}$.

$$\sqrt{x} dx = \frac{t^{-1/4}}{2 \cdot 3^{3/4}} dt$$

Read off Gamma. Here $s - 1 = -1/4$, so $s = 3/4$. A Gamma value is already an exact answer.

$$I = \boxed{\frac{\Gamma(3/4)}{2 \cdot 3^{3/4}}}$$

The master differential: $x^{p-1} dx = \frac{1}{rc^{p/r}} t^{p/r-1} dt$. This single line accounts for every power and scale.

WORKED INTEGRALS

Half-integers and interval length

Recurrence reduces half-integer Gamma values to $\sqrt{\pi}$. Rescaling a finite interval introduces a power of its length.

03. A Gaussian moment

$$I = \int_0^{\infty} x^2 e^{-x^2} dx$$

Convergence. The integrand behaves like x^2 at zero and decays rapidly at infinity.

Use the square. The square in the exponential becomes t .

$$t = x^2, \quad x^2 dx = \frac{1}{2} t^{1/2} dt$$

Identify the argument. The exponent $1/2$ corresponds to Gamma argument $3/2$.

$$I = \frac{1}{2} \Gamma(3/2)$$

Step down to one half. Use $\Gamma(1/2) = \sqrt{\pi}$. The integral over the whole real line would be twice this.

$$\Gamma(3/2) = \frac{1}{2} \Gamma(1/2), \quad I = \boxed{\frac{\sqrt{\pi}}{4}}$$

04. Rescale a finite interval

$$I = \int_0^2 x^{1/2} (2-x)^{3/2} dx$$

Convergence. Both endpoint exponents exceed -1 , so both endpoints are integrable.

Map the interval to $[0, 1]$. The limits 0 and 2 become 0 and 1 .

$$x = 2t, \quad dx = 2 dt$$

Collect the scale. The scale is $2^{1/2} \times 2^{3/2} \times 2 = 8$.

$$I = 8 \int_0^1 t^{1/2} (1-t)^{3/2} dt = 8B(3/2, 5/2)$$

Convert Beta to Gamma. The denominator is $\Gamma(a+b) = \Gamma(4) = 6$.

$$I = 8 \frac{(\sqrt{\pi}/2)(3\sqrt{\pi}/4)}{\Gamma(4)} = \boxed{\frac{\pi}{2}}$$

Interval scaling: $\int_0^L x^{a-1} (L-x)^{b-1} dx = L^{a+b-1} B(a, b)$, for $a, b, L > 0$.

WORKED INTEGRALS

Reveal the Beta pattern

A factor $1 - x^r$ suggests a power substitution on $[0, 1]$. A factor $1 + x^r$ suggests the infinite-interval Beta representation.

05. Uncover a hidden Beta integral

$$I = \int_0^1 \frac{x^2}{\sqrt{1-x^4}} dx$$

Convergence. Near 1, $1 - x^4$ is comparable to $4(1 - x)$, giving an integrable square-root singularity.

Remove the fourth power. The bounds stay 0 and 1.

$$t = x^4, \quad x^2 dx = \frac{1}{4} t^{-1/4} dt$$

Match both endpoint powers. Read $a = 3/4$ and $b = 1/2$. Both are positive.

$$I = \frac{1}{4} \int_0^1 t^{-1/4} (1-t)^{-1/2} dt$$

Give the exact value. Do not force a factorial simplification when the arguments are not integers or half-integers.

$$I = \boxed{\frac{\sqrt{\pi} \Gamma(3/4)}{4\Gamma(5/4)}}$$

06. A rational integral on the half-line

$$I = \int_0^\infty \frac{x^2}{(1+x^2)^3} dx$$

Convergence. Near zero the integrand is like x^2 ; at infinity it is like x^{-4} .

First remove the square. Match the form $t^{a-1}/(1+t)^{a+b}$.

$$t = x^2, \quad I = \frac{1}{2} \int_0^\infty \frac{t^{1/2}}{(1+t)^3} dt$$

Read both parameters. The second Beta parameter is denominator power minus the first parameter.

$$a = \frac{3}{2}, \quad a + b = 3, \quad b = \frac{3}{2}$$

Evaluate. $B(3/2, 3/2) = (\pi/4)/2 = \pi/8$; the substitution contributes another factor $1/2$.

$$I = \frac{1}{2} B(3/2, 3/2) = \boxed{\frac{\pi}{16}}$$

Match the denominator carefully. In $t^{a-1}/(1+t)^q$, the Beta parameters are a and $q - a$. Both must be positive.

WORKED INTEGRALS

Reflection and trigonometric powers

Euler's reflection identity is $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$. For the convergent Beta integral used here, $0 < z < 1$.

07. Use the reflection formula

$$I = \int_0^\infty \frac{dx}{1+x^4}$$

Convergence. The integrand stays finite at zero and behaves like x^{-4} at infinity.

Turn the power into t. The two Beta parameters must add to the denominator exponent 1.

$$t = x^4, \quad I = \frac{1}{4} \int_0^\infty \frac{t^{-3/4}}{1+t} dt$$

Identify a complementary pair. The denominator $\Gamma(1)$ equals 1.

$$I = \frac{1}{4} B(1/4, 3/4) = \frac{1}{4} \Gamma(1/4) \Gamma(3/4)$$

Reflect. Euler's reflection identity is useful because the two arguments add to 1.

$$I = \frac{\pi}{4 \sin(\pi/4)} = \boxed{\frac{\pi}{2\sqrt{2}}}$$

08. Powers of sine and cosine

$$I = \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta$$

Convergence. The endpoint exponents 4 and 2 both exceed -1.

Use $t = \sin^2 \theta$. The differential lowers both transformed powers by 1/2.

$$t = \sin^2 \theta, \quad d\theta = \frac{dt}{2\sqrt{t(1-t)}}$$

Recognize Beta. For $\sin^m \theta \cos^n \theta$, the parameters are $(m+1)/2$ and $(n+1)/2$.

$$I = \frac{1}{2} \int_0^1 t^{3/2} (1-t)^{1/2} dt = \frac{1}{2} B(5/2, 3/2)$$

Evaluate. The factor 1/2 is essential.

$$I = \frac{1}{2} \frac{(3\sqrt{\pi}/4)(\sqrt{\pi}/2)}{\Gamma(4)} = \boxed{\frac{\pi}{32}}$$

Trigonometric range: the factor $\frac{1}{2} B((m+1)/2, (n+1)/2)$ applies to $[0, \pi/2]$, with $m, n > -1$. Do not reuse it unchanged on a different interval.

WORKED INTEGRALS

Logarithms expose parameters

A logarithm can become a power under $t = -\ln x$. Alternatively, differentiate a parameter: $\partial_a t^{a-1} = t^{a-1} \ln t$.

09. A logarithm becomes a power

$$I = \int_0^1 x^2 (-\ln x)^3 dx$$

Convergence. After substitution this becomes a Gamma integral with positive shape 4 and rate 3.

Open the finite interval. As x increases from 0 to 1, t decreases from infinity to 0.

$$t = -\ln x, \quad x = e^{-t}, \quad dx = -e^{-t} dt$$

Reverse the bounds. The minus sign reverses the limits. The differential supplies one extra exponential factor.

$$I = \int_0^\infty t^3 e^{-3t} dt$$

Evaluate the scaled Gamma. In general, $x^{p-1}(-\ln x)^{q-1}$ gives $\Gamma(q)/p^q$ when $p, q > 0$.

$$I = \frac{\Gamma(4)}{3^4} = \boxed{\frac{2}{27}}$$

10. Differentiate a parameter

$$I = \int_0^1 x^2 (1-x) \ln x dx$$

Convergence. For positive Beta parameters, differentiation is justified locally by an integrable bound even with the logarithm.

Generate the logarithm. Differentiating t^{a-1} with respect to a produces $\ln t$.

$$\partial_a B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} \ln t dt$$

Differentiate the Gamma ratio. Here $\psi(z) = \Gamma'(z)/\Gamma(z)$. At positive integers, $\psi(n) = H_{n-1} - \gamma$.

$$I = B(3, 2) [\psi(3) - \psi(5)]$$

Cancel the constant. The negative sign agrees with $\ln x < 0$ on $(0, 1)$. Directly integrating $(x^2 - x^3) \ln x$ gives the same result.

$$I = \frac{1}{12} \left(-\frac{1}{3} - \frac{1}{4} \right) = \boxed{-\frac{7}{144}}$$

Why parameter differentiation is allowed: on a small neighborhood of positive a, b , endpoint powers still have exponents above -1 . Their product with $|\ln t|$ has an integrable majorant, allowing differentiation under the integral. Here $\psi(z) = \Gamma'(z)/\Gamma(z)$ and $\psi(n) = H_{n-1} - \gamma$, where $H_k = \sum_{j=1}^k 1/j$.

FORMULA REFERENCE

Match the shape, then the parameters

These identities describe ordinary convergent real integrals under the stated conditions. Each is obtained from the Gamma or Beta definition by the displayed substitution.

Integral and value	Substitution	Conditions
$\int_0^\infty x^{p-1} e^{-cx^r} dx = \frac{\Gamma(p/r)}{rc^{p/r}}$	$t = cx^r$	$p, c, r > 0$
$\int_0^L x^{a-1} (L-x)^{b-1} dx$ $= L^{a+b-1} B(a, b)$	$x = Lt$	$a, b, L > 0$
$\int_0^1 x^{p-1} (1-x^r)^{q-1} dx = \frac{1}{r} B(p/r, q)$	$t = x^r$	$p, q, r > 0$
$\int_0^\infty \frac{x^{p-1}}{(1+cx^r)^q} dx$ $= \frac{B(p/r, q-p/r)}{rc^{p/r}}$	$t = cx^r$	$c, r > 0;$ $0 < p < rq$
$\int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta$ $= \frac{1}{2} B\left(\frac{m+1}{2}, \frac{n+1}{2}\right)$	$t = \sin^2 \theta$	$m, n > -1$
$\int_0^1 x^{p-1} (-\ln x)^{q-1} dx = \frac{\Gamma(q)}{p^q}$	$t = -\ln x$	$p, q > 0$
$\int_0^\infty \frac{x^{p-1}}{1+x^r} dx = \frac{\pi}{r \sin(\pi p/r)}$	$t = x^r,$ then reflection	$r > 0;$ $0 < p < r$

Endpoint tests explain the conditions

For the rational family, the integrand is like x^{p-1} at zero and like $c^{-q}x^{p-1-rq}$ at infinity. Thus $p > 0$ and $p < rq$. At the boundary $p = rq$, the tail is comparable to $1/x$ and diverges logarithmically. For the trigonometric family, $\sin \theta \sim \theta$ near zero and $\cos \theta \sim \pi/2 - \theta$ near $\pi/2$. Therefore both exponents must exceed -1 .

Analytic continuation does not repair a divergent integral. Recurrence gives $\Gamma(-1/2) = -2\sqrt{\pi}$, but the nonnegative integral $\int_0^\infty t^{-3/2} e^{-t} dt$ diverges at zero. Its value cannot be that negative number.

From NIST DLMF §5.9.1, §5.12.1–3, and §5.5.3, with substitutions derived here.

PRACTICE

Eight integrals to recognize

For each problem: (1) check convergence, (2) name the target family, (3) transform the variable, differential and bounds, and (4) simplify. One integral diverges. Full solutions follow on pages 11–12.

01.

$$\int_0^{\infty} x^4 e^{-3x} dx$$

05.

$$\int_0^{\infty} \frac{x}{(1+x^3)^2} dx$$

02.

$$\int_0^{\infty} x^5 e^{-2x^3} dx$$

06.

$$\int_0^{\pi/2} \sin^3 \theta \cos^5 \theta d\theta$$

03.

$$\int_0^1 x^3 (1-x)^2 dx$$

07.

$$\int_0^1 x \sqrt{-\ln x} dx$$

04.

$$\int_0^3 \sqrt{x}(3-x) dx$$

08.

$$\int_0^{\infty} \frac{x^3}{(1+x^2)^2} dx$$

SOLUTIONS / 1–4

Keep track of every scale factor

1. Exponential scaling

At zero, x^4 is integrable; at infinity, exponential decay dominates. Set $t = 3x$. Then $x^4 dx = t^4 dt / 3^5$:

$$I = \frac{\Gamma(5)}{3^5} = \frac{4!}{243} = \boxed{\frac{8}{81}}.$$

2. A cubic exponential

The endpoints are integrable. Put $t = 2x^3$. The power and differential combine to $x^5 dx = t dt / 12$:

$$I = \frac{1}{12} \int_0^\infty t e^{-t} dt = \frac{\Gamma(2)}{12} = \boxed{\frac{1}{12}}.$$

Equivalently, use the master formula with $p = 6$, $r = 3$, and $c = 2$. The Gamma argument is $p/r = 2$.

3. Endpoint powers already in Beta form

The exponents 3 and 2 are positive. Read $a = 4$, $b = 3$ directly:

$$I = B(4, 3) = \frac{\Gamma(4)\Gamma(3)}{\Gamma(7)} = \frac{3!2!}{6!} = \boxed{\frac{1}{60}}.$$

4. A finite interval of length three

The endpoint exponents are $1/2$ and 1 , both above -1 . With $x = 3t$, the scale is $3^{1/2} \cdot 3 \cdot 3 = 3^{5/2}$:

$$\begin{aligned} I &= 3^{5/2} B(3/2, 2) \\ &= 3^{5/2} \frac{(\sqrt{\pi}/2) \cdot 1}{15\sqrt{\pi}/8} = 3^{5/2} \frac{4}{15} = \boxed{\frac{12\sqrt{3}}{5}}. \end{aligned}$$

An independent check for problem 4. Expanding gives $3x^{1/2} - x^{3/2}$. Its elementary integral from 0 to 3 is $2 \cdot 3^{3/2} - (2/5)3^{5/2} = 12\sqrt{3}/5$. Agreement is a useful check on the interval scale.

A five-line self-check

- Is the special-function argument one more than the transformed exponent?
- Did the substitution change the bounds or reverse their order?
- Did every scale in the integrand and the differential survive?
- Are both Beta parameters positive?
- Does the sign agree with the sign of the original integrand?

SOLUTIONS / 5–8

Parameters also test convergence

5. A rational tail

The integrand is like x at zero and x^{-5} at infinity. With $t = x^3$, $x dx = t^{-1/3} dt/3$, so

$$I = \frac{1}{3}B(2/3, 4/3) = \frac{1}{9}\Gamma(2/3)\Gamma(1/3) = \boxed{\frac{2\pi}{9\sqrt{3}}}.$$

Use recurrence $\Gamma(4/3) = \Gamma(1/3)/3$ and reflection at $1/3$.

6. Odd trigonometric powers

Both endpoints converge. With $t = \sin^2 \theta$,

$$I = \frac{1}{2}B(2, 3) = \frac{1}{2} \frac{1! 2!}{4!} = \boxed{\frac{1}{24}}.$$

7. A fractional power of a logarithm

Set $t = -\ln x$. Then $x dx = -e^{-2t} dt$, giving a convergent Gamma integral:

$$I = \int_0^\infty t^{1/2} e^{-2t} dt = \frac{\Gamma(3/2)}{2^{3/2}} = \boxed{\frac{\sqrt{\pi}}{4\sqrt{2}}}.$$

8. The integral that diverges

For large x , $x^3/(1+x^2)^2 \sim 1/x$. More concretely, for $x \geq 1$, $(1+x^2)^2 \leq 4x^4$, so the integrand is at least $1/(4x)$. The integral diverges to $+\infty$ by comparison. The formal substitution gives $\frac{1}{2}B(2, 0)$, outside the convergent Beta domain.

Sources and verification scope

The definitions and standard identities were checked against the NIST Digital Library of Mathematical Functions:

- <https://dlmf.nist.gov/5.2>: Gamma integral, analytic continuation, digamma.
- <https://dlmf.nist.gov/5.4>: factorial and half-integer values; digamma at integers.
- <https://dlmf.nist.gov/5.5>: recurrence and Euler reflection.
- <https://dlmf.nist.gov/5.9>: scaled exponential integral.
- <https://dlmf.nist.gov/5.12>: Beta definition, Gamma ratio, trigonometric and rational forms.

All 17 convergent worked/practice values were independently checked with high-precision numerical integration. The divergent exercise is established by the comparison above. Numerical checks support calculations; the stated arguments establish convergence. The companion calculator's Gamma core was checked against an independent high-precision implementation. Figures are direct samples of the stated integrands, with cropping disclosed.